

ATPL EASA STUDY GUIDE | 033 - FLIGHT PLANNING

Chapter 5: Jeppesen En-Route Charts & Airway Dimensions

Syllabus Source: CAE Oxford (pp. 165-208) | Pass Target: >= 90% AviationExam

1. AIRWAY ALTITUDES & DEFINITIONS ON JEPPESEN CHARTS

Airways on IFR en-route charts (LO and HI) are published with exact altitude constraints ensuring terrain clearance and radio reception:

Altitude Type Code	Jeppesen Representation & Meaning	Guaranteed Protection (Terrain vs Signal)
Altitude Type Code	Jeppesen Representation & Meaning	Guaranteed Protection (Terrain vs Signal)
MEA (Minimum En-route Alt)	Printed as bold number (e.g. 8000). Lowest published altitude between fixes.	Guarantees BOTH acceptable navigational signal reception AND mandatory obstacle clearance.
MOCA (Min Obstacle Clr Alt)	Printed with an asterisk * (e.g. *4500).	Guarantees obstacle clearance along entire airway, but signal reception ONLY within 22 NM of VOR!
MORA (Route MORA)	Printed with an 'a' (e.g. 7200a).	Guarantees obstacle clearance within 10 NM of airway centerline, regardless of route width.
Grid MORA	Large bold numbers inside each latitude/longitude quadrangle (e.g. 64 = 6,400 ft).	Clears all terrain/structures in the quadrangle by 1,000 ft (areas <= 5,000 ft) or by 2,000 ft in mountainous areas.
MAA (Max Authorized Alt)	Prefixed by 'MAA' (e.g. MAA FL 280).	Highest altitude allowed on the airway to prevent radio frequency interference with adjacent stations.

2. AIRWAY STRUCTURE & WAYPOINTS

- Airway Width: Standard ICAO airway corridor width is 10 NM total (5 NM on either side of centerline).
- Compulsory Reporting Point: Solid filled black triangle. Aircraft MUST report position, time, and level to ATC when crossing.
- On-Request Reporting Point: Open hollow triangle. Report position only if specifically requested by ATC.
- Changeover Point (COP): Indicates where pilots must switch NAV frequency from the behind station to the ahead station. If no COP is shown, change over occurs at the EXACT MIDPOINT of the airway leg!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Jeppesen Chart Altitudes: MEA v

Rule 1 (MEA - Minimum Enroute Altitude): - Published as a plain number (e.g. 8000). - Guarantees BOTH terrain obstacle clearance (1,000 ft flat / 2,000 ft mountain) AND continuous VHF navaid reception along the entire segment. Rule 2 (MOCA - Minimum Obstacle Clearance Altitude): - Published with a 'T' suffix (e.g. 6500T). - Guarantees obstacle clearance along the whole segment, but VHF navaid reception ONLY within 22 NM of the VOR!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Reading En-Route Altitudes & Frequency Change**

SCENARIO (Standard En-Route Chart Reading Drill): A pilot is planning an IFR flight along airway V12 between VOR Alfa and VOR Bravo: - Total distance between VORs = 90 NM - Along the airway segment, the chart displays: '7000' and '*4200' - No Changeover Point (COP) symbol is depicted along the route QUESTION 1: What do the numbers '7000' and '*4200' mean? QUESTION 2: At what distance from VOR Alfa must the pilot switch navigation receivers to VOR Bravo? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Decode '7000' and '*4200': - '7000' without an asterisk is the MEA (Minimum En-route Altitude) = 7,000 ft. - '*4200' with an asterisk is the MOCA (Minimum Obstacle Clearance Altitude) = 4,200 ft. - Operational Meaning: To fly legally with reliable radio navigation, you must cruise at or above 7,000 ft (MEA). If descending in an emergency, you are guaranteed terrain clearance down to 4,200 ft (MOCA), but VOR signal is only guaranteed within 22 NM of the VOR! Step 2: Determine the Changeover Point (COP): - Rule: When no COP symbol is published, the frequency changeover MUST take place at the EXACT MIDPOINT of the leg! - Total distance = 90 NM - Midpoint = $90 \text{ NM} / 2 = 45 \text{ NM}$ from VOR Alfa! FINAL ANSWER: 7000 = MEA; *4200 = MOCA. Switch navigation frequency to VOR Bravo at exactly 45 NM from VOR Alfa.

[!] EXAM TRAP (AVIATIONEXAM) - The MOCA 22 NM Signal Reception Trap

- MOCA guarantees OBSTACLE CLEARANCE along the entire airway. - But MOCA guarantees NAVIGATION SIGNAL RECEPTION only within 22 NAUTICAL MILES (40 km) of the VOR transmitter! Beyond 22 NM, you may lose the VOR radial!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: On a Jeppesen en-route chart, what does an altitude preceded by an ASTERISK (*) (e.g. *3500) signify?

- [A]: Maximum Authorized Altitude
- [B]: MOCA (Minimum Obstacle Clearance Altitude)
- [C]: Transition altitude
- [D]: Minimum crossing altitude

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The asterisk denotes MOCA. It guarantees terrain clearance along the entire segment, but navigational signal coverage only within 22 NM of the facility.

Q2: In the absence of a published Changeover Point (COP) on an airway between two VORs, where should the

- [A]: At 20 NM from the first VOR
- [B]: At the EXACT MIDPOINT between the two facilities

- [C]: Only upon ATC instruction
- [D]: Over the next reporting point

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. By standard ICAO rules, if no COP is shown on the chart, the frequency changeover must be performed halfway between the two navigational aids.

Q3: In a Grid MORA quadrangle over mountainous terrain, what vertical clearance above the highest obstacle is required?

- [A]: 1,000 ft
- [B]: At least 2,000 FT vertically
- [C]: 500 ft
- [D]: 35 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Grid MORA provides 1,000 ft clearance in non-mountainous terrain, and at least 2,000 ft clearance in designated mountainous areas (terrain elevation above 5,000 ft).